

Creating OSIVQ cognitive style clusters

This short vignette details how the OSIVQ clusters were created and named, and how they relate to the VVIQ-based groups.

```
library(aphantasiaReasoningViie)
#> Welcome to aphantasiaReasoningViie.
```

First, let's get the cleaned, analysis-ready data (see `vignette("preparing_data")` for details).

```
df_survey <- get_clean_data()$df_survey
```

The OSIVQ clusters were created using a consensus of three clustering algorithms (GMM, PAM and C-Means) applied to the three OSIVQ sub-scales (Object, Spatial and Verbal). A function was created for this specific task, `cluster_osivq()`, which uses the `diceR::dice()` function from the *diceR* package under the hood. Following the methodology proposed by Delem et al. (2025), we searched for three “visualiser”, “spatialiser” and “verbaliser” clusters based on their most “dominant” OSIVQ sub-scale score.

However, the clustering algorithm does not name the clusters, so we needed to check their properties to assign meaningful names. The function `add_named_clusters()` adds the cluster assignments to the data frame, by default with generic names. `summarise_clustering()` then provides a summary of the clusters' properties, which allowed us to choose meaningful names for the three clusters produced.

```
# Clustering OSIVQ data
clustering <- cluster_osivq(df_survey)

# Checking cluster properties to define names for each cluster
df_survey |>
  add_named_clusters(clustering) |>
  summarise_clustering()
#> # A tibble: 9 x 8
#>   group      cluster      n vviq object spatial verbal raven
#>   <fct>      <fct>    <int> <dbl> <dbl>    <dbl> <dbl> <dbl>
#> 1 Hypophantasia cluster_1      3  24    1.67    3.67    2.55  15.7
#> 2 Typical      cluster_1     12 50.9    2.39    2.84    2.46  15.3
#> 3 Aphantasia   cluster_1      4  16    1.41    3.32    1.97   17
#> 4 Typical      cluster_2     38 57.3    3.71    2.87    3.09  15.8
#> 5 Hyperphantasia cluster_2      4 77.5    4.41    3.93    3.56   15
#> 6 Aphantasia   cluster_2      1  16    4.93    3.27    4.44   16
#> 7 Aphantasia   cluster_3     25 16.0    1.25    2.26    3.38  16.0
#> 8 Hypophantasia cluster_3     14 23.7    1.36    2.52    3.5   15.3
#> 9 Typical      cluster_3      3 50.3    1.76    2.67    4.11  17.3
```

We saw here that cluster 1 was the spatialiser one, cluster 2 was the visualiser and cluster 3 the verbaliser. We can now use `add_named_clusters()` again with its optional arguments to add these labels, reorder them, choose factor levels and add contrasts for planned comparisons between the visualiser cluster and the other two.

```
# Adding named clusters to the survey data
df_survey <- add_named_clusters(
  df_survey, clustering,
  names = c("Spatialiser", "Visualiser", "Verbaliser"),
```

```

levels = c("Visualiser", "Spatialiser", "Verbaliser"),
contrasts = c("_visualiser", "_spatialiser", "_verbaliser"),
base = 1
)

# Let's check the cluster properties again
contrasts(df_survey$cluster)
#>      _spatialiser _verbaliser
#> Visualiser      0           0
#> Spatialiser     1           0
#> Verbaliser      0           1
summarise_clustering(df_survey)
#> # A tibble: 9 x 8
#>   group      cluster      n vviq object spatial verbal raven
#>   <fct>      <fct>    <int> <dbl> <dbl> <dbl> <dbl> <dbl>
#> 1 Typical    Visualiser    38  57.3  3.71  2.87  3.09  15.8
#> 2 Hyperphantasia Visualiser     4  77.5  4.41  3.93  3.56  15
#> 3 Aphantasia  Visualiser     1  16    4.93  3.27  4.44  16
#> 4 Hypophantasia Spatialiser     3  24    1.67  3.67  2.55  15.7
#> 5 Typical    Spatialiser    12  50.9  2.39  2.84  2.46  15.3
#> 6 Aphantasia Spatialiser     4  16    1.41  3.32  1.97  17
#> 7 Aphantasia Verbaliser    25  16.0  1.25  2.26  3.38  16.0
#> 8 Hypophantasia Verbaliser    14  23.7  1.36  2.52  3.5   15.3
#> 9 Typical    Verbaliser     3  50.3  1.76  2.67  4.11  17.3

```

`cluster_osivq()` uses a seed, so the resulting clusters will always be the same for a given seed and data set. Thus, we were able to use this short clustering pipeline in all subsequent analyses to create our “cluster” variable and add it to the experimental data set for modelling.

```

#> - Session info -----
#> setting value
#> version R version 4.5.0 (2025-04-11 ucrt)
#> os      Windows 11 x64 (build 26100)
#> system  x86_64, mingw32
#> ui      RTerm
#> language (EN)
#> collate French_France.utf8
#> ctype   French_France.utf8
#> tz      Europe/Paris
#> date    2025-09-17
#> pandoc  3.6.3 @ C:/Program Files/RStudio/resources/app/bin/quarto/bin/tools/ (via rmarkdown)
#>
#> - Packages -----
#> ! package          * version      date (UTC) lib source
#> P abind             1.4-8        2024-09-12 [?] CRAN (R 4.5.0)
#>   aphantasiaReasoningViie * 0.0.0.9000  2025-09-16 [1] local
#> P assertthat        0.2.1        2019-03-21 [?] CRAN (R 4.5.0)
#> P class              7.3-23       2025-01-01 [3] CRAN (R 4.5.0)
#> P cli                3.6.5        2025-04-23 [?] CRAN (R 4.5.0)
#> P clue               0.3-66       2024-11-13 [?] CRAN (R 4.5.0)
#> P cluster            2.1.8.1      2025-03-12 [3] CRAN (R 4.5.0)
#> P clusterCrit        1.3.0        2023-11-23 [?] CRAN (R 4.5.0)
#> P clValid            0.7          2021-02-14 [?] CRAN (R 4.5.0)
#> P combinat           0.0-8        2012-10-29 [?] CRAN (R 4.5.0)
#> P crayon             1.5.3        2024-06-20 [?] CRAN (R 4.5.0)
#> P diceR              3.1.0        2025-06-19 [?] RSPM (R 4.5.0)
#> P digest             0.6.37       2024-08-19 [?] CRAN (R 4.5.0)
#> P dplyr              1.1.4        2023-11-17 [?] CRAN (R 4.5.0)
#> P e1071              1.7-16       2024-09-16 [?] CRAN (R 4.5.0)
#> P evaluate           1.0.4        2025-06-18 [?] RSPM (R 4.5.0)
#> P farver             2.1.2        2024-05-13 [?] CRAN (R 4.5.0)
#> P fastmap            1.2.0        2024-05-15 [?] CRAN (R 4.5.0)
#> P forcats            1.0.0        2023-01-29 [?] CRAN (R 4.5.0)
#> P generics           0.1.4        2025-05-09 [?] CRAN (R 4.5.0)
#> P ggplot2            3.5.2        2025-04-09 [?] CRAN (R 4.5.0)
#> P glue               1.8.0        2024-09-30 [?] CRAN (R 4.5.0)
#> P gtable             0.3.6        2024-10-25 [?] CRAN (R 4.5.0)
#> P haven              2.5.5        2025-05-30 [?] RSPM (R 4.5.0)
#> P highr              0.11         2024-05-26 [?] CRAN (R 4.5.0)
#> P hms                1.1.3        2023-03-21 [?] CRAN (R 4.5.0)
#> P htmltools          0.5.8.1      2024-04-04 [?] CRAN (R 4.5.0)
#> P httpuv             1.6.16       2025-04-16 [?] CRAN (R 4.5.0)
#> P klaR               1.7-3        2023-12-13 [?] CRAN (R 4.5.0)
#> P knitr              1.50         2025-03-16 [?] CRAN (R 4.5.0)
#> P labelled           2.14.1       2025-05-06 [?] CRAN (R 4.5.0)
#> P later              1.4.2        2025-04-08 [?] CRAN (R 4.5.0)
#> P lifecycle          1.0.4        2023-11-07 [?] CRAN (R 4.5.0)
#> P magrittr           2.0.3        2022-03-30 [?] CRAN (R 4.5.0)
#> P MASS               7.3-65       2025-02-28 [3] CRAN (R 4.5.0)
#> P mclust             6.1.1        2024-04-29 [?] CRAN (R 4.5.0)
#> P mime               0.13         2025-03-17 [?] CRAN (R 4.5.0)
#> P miniUI             0.1.2        2025-04-17 [?] CRAN (R 4.5.0)
#> P pillar             1.10.2       2025-04-05 [?] CRAN (R 4.5.0)

```

```

#> P pkgconfig          2.0.3      2019-09-22 [?] CRAN (R 4.5.0)
#> P promises           1.3.3      2025-05-29 [?] RSPM
#> P proxy              0.4-27     2022-06-09 [?] CRAN (R 4.5.0)
#> P purrr             1.0.4      2025-02-05 [?] CRAN (R 4.5.0)
#> P questionr         0.8.1      2025-06-10 [?] RSPM (R 4.5.0)
#> P R6                 2.6.1      2025-02-15 [?] CRAN (R 4.5.0)
#> P RColorBrewer       1.1-3      2022-04-03 [?] CRAN (R 4.5.0)
#> P Rcpp              1.0.14     2025-01-12 [?] CRAN (R 4.5.0)
#> P rlang              1.1.6      2025-04-11 [?] CRAN (R 4.5.0)
#> P rmarkdown         2.29       2024-11-04 [?] CRAN (R 4.5.0)
#> P rstudioapi        0.17.1     2024-10-22 [?] CRAN (R 4.5.0)
#> P scales            1.4.0      2025-04-24 [?] CRAN (R 4.5.0)
#> P sessioninfo       1.2.2      2021-12-06 [?] RSPM (R 4.5.0)
#> P shiny             1.10.0     2024-12-14 [?] CRAN (R 4.5.0)
#> P stringi           1.8.7      2025-03-27 [?] CRAN (R 4.5.0)
#> P stringr           1.5.1      2023-11-14 [?] CRAN (R 4.5.0)
#> P tibble            3.3.0      2025-06-08 [?] RSPM
#> P tidyselect        1.2.1      2024-03-11 [?] CRAN (R 4.5.0)
#> P utf8              1.2.6      2025-06-08 [?] RSPM
#> P vctrs             0.6.5      2023-12-01 [?] CRAN (R 4.5.0)
#> P withr             3.0.2      2024-10-28 [?] CRAN (R 4.5.0)
#> P xfun              0.52       2025-04-02 [?] CRAN (R 4.5.0)
#> P xtable            1.8-4      2019-04-21 [?] CRAN (R 4.5.0)
#> P yaml              2.3.10     2024-07-26 [?] CRAN (R 4.5.0)
#>
#> [1] C:/Users/Mael/AppData/Local/R/cache/R/renv/library/aphantasiaReasoningViie-d318fbb0/windows/R-4
#> [2] C:/Users/Mael/AppData/Local/R/cache/R/renv/sandbox/windows/R-4.5/x86_64-w64-mingw32/9bc714e4
#> [3] C:/Program Files/R/R-4.5.0/library
#>
#> P -- Loaded and on-disk path mismatch.
#>
#> -----

```

References

Delem, M., Turkben, S., Cousineau, D., Cavalli, E., & Plancher, G. (2025). *Unsupervised clustering reveals spatial and verbal cognitive profiles in aphantasia and typical imagery.*